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Inventor(s)	Rogers; Jon-Paul et al.

Healicoil knotless distal tip and plug transmission

Abstract

An anchor assembly can include an anchor comprising a proximal portion, a distal portion having a tip, and a surface extending between the tip and the proximal portion and can include a protrusion extending along a length of the surface. The anchor can define a cavity therein with a wall of the cavity having threads, an opening to the cavity located in the proximal anchor portion, and a through hole defined in a distal portion of the tip. Further, the anchor assembly can include a plug having a threaded proximal outer portion and a non-threaded distal outer portion disposed within the cavity of the anchor. The threaded proximal outer portion can be engaged with the cavity wall threads and the non-threaded distal outer portion can be configured to move axially within the anchor cavity. The anchor assembly can include a sleeve having an opening to receive the protrusion.

Inventors: **Rogers; Jon-Paul (North Smithfield, RI), Housman; Mark Edwin (North Attleborough, MA), Rego, Jr.; Richard Perry (Mansfield, MA), Cashmore; Peter James (Durham, NH)**

Applicant: **Smith & Nephew, Inc.** (Memphis, TN); **Smith & Nephew Orthopaedics AG** (Zug, CH); **Smith & Nephew Asia Pacific Pte. Limited** (Singapore, SG)

Family ID: **72811937**

Assignee: **SMITH & NEPHEW, INC. (Memphis, TN); SMITH & NEPHEW ORTHOPAEDICS AG (Zug, CH); SMITH & NEPHEW ASIA PACIFIC PTE. LIMITED (Singapore, SG)**

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Primary Examiner: Labranche; Brooke

Attorney, Agent or Firm: ArentFox Schiff LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority to U.S. Provisional Application No. 62/928,576, entitled “HEALICOIL KNOTLESS DISTAL TIP AND PLUG TRANSMISSION,” filed on Oct. 31, 2019, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD
(1) The described technology relates generally to tissue repair, and more specifically, to an anchor for securing tissue to bone.
BACKGROUND
(2) Arthroscopic procedures often require soft tissue to be reattached to bone. To achieve this, anchors are placed in the bone and sutures attached to the anchor are passed through the tissue to securely retain the tissue in place. When making a repair of soft tissue to bone, it is advantageous to have as large an area of contact between the bone and tissue as possible. Anchor points spaced from one another in rows result in a repair having a broader area of contact.
SUMMARY
(3) A procedure, and components for use in such procedure, that securely attaches tissue to bone using a plurality of attachment points over a large area of contact is needed. Such procedure must be able to be done in a quick and efficient manner with a minimum of recovery time for the patient.
(4) In one aspect, the present disclosure relates to an anchor assembly for securing tissue to bone. The anchor assembly can include an anchor having a proximal portion, a distal portion having a tip, and a surface extending between the tip and the proximal portion. The surface can include a protrusion extending along a length of the surface. The anchor can define a cavity therein with a wall of the cavity having threads. Further, the anchor can have an opening to the cavity located in the proximal anchor portion and a through hole defined in a distal portion of the tip. Further, the anchor assembly can include a plug having a threaded proximal outer portion and a non-threaded distal outer portion disposed within the cavity of the anchor. The threaded proximal outer portion can be engaged with the cavity wall threads and the non-threaded distal outer portion can be configured to move axially within the anchor cavity. Further, the anchor assembly can include a sleeve having an opening to receive the protrusion. The sleeve can be releasably coupled to the anchor and disposed about the surface at the proximal portion of the anchor. Further, the protrusion and the opening are configured to reduce the rotation of the anchor with respect to the sleeve.
(5) In some embodiments, the plug can include an internal cavity having a proximal inner portion. The proximal inner portion can include at least one of a D-shape, a triangle shape, a rectangle

shape, a square shape, and a star shape.

(6) In some embodiments, the anchor includes a plurality of protrusions extending along a length of the surface. In some embodiments, the sleeve can include a plurality of openings. Each opening is configured to receive the respective plurality of protrusions. The sleeve can include an internal cavity having a distal inner portion. The distal inner portion has at least one of a D-shape, a triangle shape, a rectangle shape, a square shape, and a star shape. In some embodiments, the sleeve can include an internal cavity having a threaded distal inner portion. The threaded distal inner portion can have a similar shape as the proximal outer portion of the plug.

(7) In one aspect the present disclosure relates to a method of tissue repair. The method can include driving an anchor assembly of an anchor system into a bone hole with an anchor driver. The anchor system can include a sleeve. The sleeve can include at least one open helical coil having a proximal end and a distal end wherein the at least one open helical coil defines an internal volume communicating with a region exterior to the at least one open helical coil through a spacing between turns of the at least one open helical coil. The sleeve can further include at least one rib disposed within the internal volume, connected to at least two turns of the at least one open helical coil, wherein the at least one rib is engageable with a grooved outer shaft of the anchor driver. The anchor assembly is engageable with the sleeve. The anchor assembly can include an anchor comprising a proximal portion, a distal portion having a tip, and a surface extending between the tip and the proximal portion, the surface including a protrusion extending along a length of the surface, the anchor defining a cavity therein, with a wall of the cavity having threads, and the anchor having an opening to the cavity located in the proximal anchor portion and a through hole defined in a distal portion of the tip. The anchor assembly can also include a plug, having a threaded proximal outer portion and a non-threaded distal outer portion, disposed within the cavity of the anchor, wherein the threaded proximal outer portion is engaged with the cavity wall threads and the non-threaded distal outer portion is configured to move axially within the anchor cavity. The sleeve has an opening to receive the protrusion, releasably coupled to the anchor and disposed about the surface at the proximal portion of the anchor, wherein the protrusion and the opening are configured to reduce the rotation of the anchor with respect to the sleeve.

(8) The method can further include advancing the sleeve into the bone and into engagement with the anchor assembly using the anchor driver. The anchor driver further includes an axially compliant member configured to allow a relative motion between the grooved outer shaft and an inner shaft along a longitudinal axis of the anchor driver. The axially compliant member can be separate and distinct from a spring of the torque and/or travel limiter.

(9) The method can further include advancing, using the anchor driver, the plug into a distal suture-locked position to capture the suture in a locked position. The method can also include a retracting, using the anchor driver, the plug into a proximal, suture unlocked position to release the suture into a freely slidable state.

(10) In some embodiments, the method includes engaging the at least one rib of the sleeve with the grooved outer shaft of the anchor drive, engaging the suture capture member of the tip structure with the inner shaft of the anchor driver, and releasably attaching the tip structure to an intermediate shaft of the anchor driver positioned between the grooved outer shaft and the inner shaft.

(11) The method can include exerting a threshold force along a longitudinal axis of the anchor driver, causing a retraction of the grooved outer shaft of the anchor driver. Further, the method can include advancing the sleeve further comprises axially sliding the sleeve along a longitudinal axis of the anchor driver. The anchor driver further can include a housing defining a threaded internal cavity coupled with the grooved outer shaft for providing a threaded travel of the sleeve relative to the inner shaft, wherein advancing the sleeve is achieved by rotating the housing. The housing can be slidable relative to the inner shaft along the longitudinal axis of the anchor driver for providing a slidable travel of the sleeve.

(12) In some embodiments, the slidable travel of the sleeve is equal to or greater than an axial clearance defined along the longitudinal axis of the anchor driver between a surface of the bone and a distal end of the sleeve, the sleeve being substantially engaged with the grooved outer shaft. In other embodiments, a combined travel of the sleeve includes the slidable travel of the sleeve added to the threaded travel of the sleeve, is equal to or greater than the axial clearance.
(13) In some embodiments, at least one of the sleeve, the housing, or the grooved outer shaft further includes one or more locking mechanisms for preventing a longitudinal sliding motion of the housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Various aspects of at least one embodiment of the present disclosure are discussed below with reference to the accompanying figures. It will be appreciated that for simplicity and clarity of illustration, elements shown in the drawings have not necessarily been drawn accurately or to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity or several physical components may be included in one functional block or element. Further, where considered appropriate, reference numerals may be repeated among the drawings to indicate corresponding or analogous elements. For purposes of clarity, not every component may be labeled in every drawing. The figures are provided for the purposes of illustration and explanation and are not intended as a definition of the limits of the invention.
(2) FIGS. **1**A-**1**B illustrate an anchor assembly, according to certain embodiments;
(3) FIG. **2** illustrates an anchor of the anchor assembly, according to certain embodiments;
(4) FIG. **3**A-**3**B illustrate an anchor assembly, according to certain embodiments;
(5) FIG. **4** illustrates an anchor of the anchor assembly of FIGS. **3**A-**3**B, according to certain embodiments;
(6) FIG. **5** illustrates an anchor of the anchor assembly, according to certain embodiments; and
(7) FIG. **6** illustrates cross sectional surface of an internal cavity of a plug of the anchor assembly, according to certain embodiments.
DETAILED DESCRIPTION
(8) In the following detailed description, numerous specific details are set forth in order to provide a thorough understanding of the embodiments of the present disclosure. It will be understood by those of ordinary skill in the art that these embodiments may be practiced without some of these specific details. In other instances, well-known methods, procedures, components and structures may not have been described in detail so as not to obscure the described embodiments.
(9) Prior to describing at least one embodiment in detail, it is to be understood that the claims are not limited in their application to the details of construction and the arrangement of the components set forth in the following description or illustrated in the drawings. Also, it is to be understood that the phraseology and terminology employed herein are for the purpose of description only and should not be regarded as limiting.
(10) During an arthroscopic procedure, an implant is delivered to a bone hole to secure soft tissue. The present disclosure describes an assembly which consists of a two piece implant, a distal tip and a proximal body. The distal tip can be secured to an inserter middle shaft described in U.S. Pat. No. 9,526,488, incorporated herein by reference. The distal tip can include sutures which pass through an eyelet and is inserted into a prepped or no-prepped hole before delivering the proximal body. The distal plug allows for torsional loads which will allow for the suture to be adequately captured in the eyelet.
(11) The present disclosure provides an anchor assembly for securing tissue to bone. The anchor assembly provides a transmission feature between the distal plug and inner shaft which can provide

adequate torsional strength to capture the suture.
(12) FIG. **1**A illustrates an anchor assembly **100** being disassembled, according to certain embodiments. FIG. **1**B illustrates the anchor assembly **100** in assembled position, according to certain embodiments, for delivering to a bone hole. The anchor assembly **100** includes an anchor **102**. The anchor **102** has a proximal portion **106** and a distal portion **104**. The distal portion **104** includes a tip **110**. The distal end of the tip **110** may be blunt, as shown, or may be pointed. The tip **110** can be constructed from, for example but not limited to, polymers (e.g., PEEK), bioabsorbable materials, metals (e.g., surgical steel, titanium), or any other suitable material. The anchor **102** also includes a surface **108** extending between the tip **110** and the proximal portion **106**. The surface **108** includes a protrusion **120** extending along a length of the surface **108**. The protrusion **120** can have any geometrical shape. In some embodiments, the protrusion **120** has a semi-oval shape. Further, the anchor **102** defines a cavity **122** therein, with a wall of the cavity **122** having threads **124**. The anchor **102** includes an opening **126** to the cavity **122** located in the proximal anchor portion **106**. The anchor **102** can include a through hole **128** defined in a distal portion **104** of the tip **110**.
(13) As FIGS. **1**A-**1**B illustrate, the anchor assembly **100** includes a plug **340** (FIG. **4**) having a threaded proximal outer portion **342** and a non-threaded distal outer portion **344**, disposed within the cavity **122** (FIG. **2**) of the anchor **102**. The threaded proximal outer portion **342** is engaged with the cavity wall threads **124** and the non-threaded distal outer portion **344** is configured to move axially within the anchor cavity **122**.
(14) Further, the anchor assembly **100** can include a sleeve **132** having an opening **134** for receiving the protrusion **120**, as shown in FIGS. **1**A-**1**B. The sleeve **132** may have an open helical coil configuration, as described in U.S. Pat. No. 9,526,488, incorporated herein by reference. The sleeve **132** can be constructed from, for example but not limited to, polymers (e.g., PEEK), bioabsorbable materials, metals (e.g., surgical steel, titanium), or any other suitable material. The opening **134** can be releasably coupled to the anchor **102** and disposed about the surface **108** at the proximal portion **106** of the anchor **102**. The sleeve **132** includes an internal cavity **136** having a distal inner portion **138**. The distal inner portion **138** of the sleeve **132** includes a threaded surface **139**. In some embodiments, the threaded surface **139** has the same or similar shape as the proximal portion **106** of the anchor **102**. The protrusion **120** and the opening **134** are configured to reduce the rotation of the anchor **102** with respect to the sleeve **132**.
(15) FIG. **2** illustrates an anchor assembly **200**, according to certain embodiments. Similar to anchor assembly **100**, the anchor assembly **200** includes an anchor **102**. The anchor **102** has a proximal portion **106** and a distal portion **104**. The distal portion **104** includes a tip **110**. The anchor **102** also includes a surface **108** extending between the tip **110** and the proximal portion **106**. The surface **108** includes a protrusion **120** extending along a length of the surface **108**. Further, the anchor **102** defines a cavity **122** therein, with a wall of the cavity **122** having threads **124**. The anchor **102** includes an opening **126** to the cavity **122** located in the proximal anchor portion **106**. The anchor **102** can include a through hole **128** defined in a distal portion **104** of the tip **110**.
(16) As FIG. **2** illustrates the anchor assembly **200** includes a plug **340** (FIG. **4**) having a threaded proximal outer portion **342** and a non-threaded distal outer portion **344**, disposed within the cavity **122** of the anchor **102**. The threaded proximal outer portion **342** is engaged with the cavity wall threads **124** and the non-threaded distal outer portion **344** is configured to move axially within the anchor cavity **122**.
(17) The anchor assembly **200** can include a sleeve **232** having an opening **234** for receiving the protrusion **120**, as shown in FIG. **2**. The opening **234** can be releasably coupled to the anchor **102** and disposed about the surface **108** at the proximal portion **106** of the anchor **102**. The sleeve **232** includes an internal cavity **236** having a distal inner portion **238**. The distal inner portion **238** of the sleeve **232** includes the same or similar shape as the proximal portion **106** of the anchor **102**. As explained above, the protrusion **120** and the opening **234** are configured to reduce the rotation of

the anchor **102** with respect to the sleeve **232**.

(18) FIGS. **3**A-**3**B illustrate an anchor assembly **300**, according to certain embodiments. FIG. **3**A illustrates an anchor assembly **300** being disassembled, according to certain embodiments. FIG. **3**B illustrates the anchor assembly **300** in assembled position, according to certain embodiments. FIG. **4** illustrates an anchor **302** of the anchor assembly **300**, according to certain embodiments. The anchor assembly **300** includes an anchor **302**. The anchor **302** has a proximal portion **306**, a distal portion **304**, and a middle portion **305**. The distal portion **304** includes a tip **310**. The distal end of the tip **310** may be blunt, as shown in FIG. **3**A, or may be pointed as shown in FIG. **4**. The anchor **302** also includes a first surface **308** extending between the tip **310** and the middle portion **305**. Further, the anchor **302** includes a second surface **301** extending from the middle portion **305** to the proximal portion **306**. The second surface **301** includes a plurality of protrusions **320** extending along a length of the second surface **301** from the middle portion **305** to the proximal portion **306**.

(19) In some embodiments, the second surface **301** includes four protrusions **320**. The plurality of protrusions **320** can have any geometrical shape. In some embodiments, the plurality of protrusions **320** has a rectangular shape. Further, the anchor **302** defines a cavity **322** therein having an internal axis extending from the distal portion **304** to the proximal portion **306**. The second surface **301** is located closer to the internal axis than the first surface **308**. A wall of the cavity **322** has threads **324**. The anchor **302** includes an opening **326** to the cavity **322** located in the proximal anchor portion **306**. The anchor **302** can include a through hole **328** defined in a distal portion **304** of the tip **310**.

(20) The anchor assembly **300** can include a plug **340** having a threaded proximal outer portion **342** disposed within the cavity **322** of the anchor **302**. The threaded proximal outer portion **342** is engaged with the cavity wall threads **324**.

(21) Further, the anchor assembly **300** can include a sleeve **332** having an internal cavity **336** with a distal inner portion **338** having an inner surface **339**. The inner surface **339** includes a plurality of openings **334** for receiving the plurality of protrusions **320**, as shown in FIGS. **3**A-**3**B. Each opening of the plurality of openings **334** receives the respective protrusion of the plurality of protrusions **320**. The plurality of openings **334** can be releasably coupled to the anchor **302** and disposed about the second surface **301** at the middle portion **305** of the anchor **302**. The plurality of protrusions **320** and the plurality of openings **334** are configured to reduce the rotation of the anchor **302** with respect to the sleeve **332**.

(22) FIG. **5** illustrates an anchor **502** having a tip **510** and a plug **540**, according to certain embodiments. The distal end of the tip **510** may be blunt, as shown, or may be pointed. As FIG. **5** illustrates, the plug **540** includes an internal cavity **560** having a proximal inner portion **562**. The proximal inner portion **562** includes a D-shape surface. In some embodiments, the proximal inner portion **562** of the internal cavity **560** can include a triangle shape, a rectangle shape, a square shape, and a star shape (shown in FIG. **6**).

(23) It is appreciated that certain features of the invention, which are, for clarity, described in the context of separate embodiments, may also be provided in combination in a single embodiment. Conversely, various features of the invention, which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable sub-combination.

(24) Whereas many alterations and modifications of the disclosure will no doubt become apparent to a person of ordinary skill in the art after having read the foregoing description, it is to be understood that the particular embodiments shown and described by way of illustration are in no way intended to be considered limiting. Further, the subject matter has been described with reference to particular embodiments, but variations within the spirit and scope of the disclosure will occur to those skilled in the art. It is noted that the foregoing examples have been provided merely for the purpose of explanation and are in no way to be construed as limiting of the present disclosure.

(25) Although the present disclosure has been described herein with reference to particular

embodiments, the present disclosure is not intended to be limited to the particulars disclosed herein; rather, the present disclosure extends to all functionally equivalent structures, methods and uses, such as are within the scope of the claims.

Claims

1. An anchor assembly comprising: an anchor comprising a proximal portion and a distal portion, the distal portion having a tip defining a through hole extending through a width of the anchor for receiving a suture therethrough, an outer lateral surface of the proximal portion including a protrusion defined by a circumferential portion, and a single elongate portion extending from the circumferential portion towards a proximal end of the anchor, a surface of both the circumferential portion and the single elongate portion being raised relative to an entirety of a remainder of the outer lateral surface of the proximal portion, the anchor defining a cavity therein, a wall of the cavity having threads, the anchor having an opening to the cavity located in the proximal portion; a plug having a threaded proximal outer portion and a non-threaded distal outer portion, the plug disposed within the cavity of the anchor such that the threaded proximal outer portion is engaged with the cavity wall threads and the non-threaded distal outer portion is configured to move axially within the anchor cavity; and a sleeve having an opening to engage the single elongate portion, the sleeve releasably coupled to the anchor and disposed about the outer lateral surface at the proximal portion of the anchor when the sleeve is engaged with the single elongate portion, the engagement of the sleeve with the single elongate portion preventing rotation of the sleeve relative to the anchor in both a clockwise and counterclockwise direction; wherein the single elongate portion and the opening are configured to reduce the rotation of the anchor with respect to the sleeve; wherein the entirety of the remainder of the outer lateral surface of the proximal portion comprises a majority of the outer lateral surface of the proximal portion; wherein a proximal end of the single elongate portion is distal to a proximal-most end of the remainder of the outer lateral surface of the proximal portion of the anchor; and wherein the through hole is distal to the circumferential portion of the protrusion.
2. The anchor assembly of claim 1, wherein the plug comprises an internal cavity having a proximal inner portion, the proximal inner portion comprises at least one of a D-shape, a triangle shape, a rectangle shape, a square shape, and a star shape.
3. The anchor assembly of claim 1, wherein the sleeve comprises an internal cavity having a distal inner portion, the distal inner portion comprises at least one of a D-shape, a triangle shape, a rectangle shape, a square shape, and a star shape.
4. The anchor assembly of claim 1, wherein the sleeve comprises an internal cavity having a threaded distal inner portion, the threaded distal inner portion having a similar shape as the proximal outer portion of the plug.
5. The anchor assembly of claim 1, wherein a distal end of the tip is blunt.
6. The anchor assembly of claim 1, wherein a distal end of the tip is pointed.
7. The anchor assembly of claim 1, wherein the single elongate portion has a semi-oval shape.
8. The anchor assembly of claim 1, wherein at least one of the tip and the sleeve comprises polymers, bioabsorbable materials, or metals.
9. The anchor assembly of claim 1, wherein the protrusion is non-threaded.